Demand Response Program

Database Schema — Equipment & Enrollment Management

Version 2.0 | Includes SkySpark Integration

This schema normalizes the pipeline spreadsheet into a relational structure optimized for filtering by region, ISO, utility, client, equipment type, and DR program. It has been updated to include SkySpark integration tables for live telemetry and event performance tracking. Primary keys (PK) are shown in blue; foreign keys (FK) are shown in teal.

PK	Primary Key
FK	Foreign Key

Reference & Lookup Tables

CLIENT

One row per client organization. Centralizes client name and type to eliminate duplication across sites.

Column	Type	Key	Notes
client_id	uuid	PK	Unique identifier for each client
client_name	varchar		e.g. Advocate Health, Caltech
client_type	varchar		e.g. Health Care, Higher Ed — use a controlled vocabulary

REGION

Geographic regions used to group sites for filtering. Can represent states, utility territories, or custom regions.

Column	Type	Key	Notes
region_id	uuid	PK	Unique identifier for each region
region_name	varchar		e.g. Midwest, Southeast, Texas
state	varchar		Two-letter state abbreviation (nullable for multi-state regions)

ISO

Independent System Operators. Stored as a lookup to avoid concatenated strings (e.g. 'MISO/NA/SPPISO') in source data.

Column	Type	Key	Notes
iso_id	uuid	PK	Unique identifier for the ISO
iso_name	varchar		e.g. ERCOT, CAISO, PJM, MISO, ISONE, NYISO, SPPISO, TVA, N/A

UTILITY

Utilities and retail electricity suppliers. Linked to a region for geographic filtering.

Column	Type	Key	Notes
utility_id	uuid	PK	Unique identifier for the utility
utility_name	varchar		e.g. PG&E;, TXU Energy, Duke Indiana, LADWP
region_id	uuid	FK	Region this utility primarily serves

MARKET_PARTICIPANT

Aggregators and curtailment service providers (CSPs) that enroll equipment in DR programs on the client's behalf.

Column	Type	Key	Notes
mp_id	uuid	PK	Unique identifier for the market participant
mp_name	varchar		e.g. CPower, ERock, Honeywell, Duke

DR_PROGRAM

Demand response programs offered by ISOs or utilities. A site's equipment can be enrolled in one or more programs.

Column	Type	Key	Notes
program_id	uuid	PK	Unique identifier for the program
program_name	varchar		e.g. 4CP, ERS, LMR, DSGS, ADCR
program_stack	varchar		Combined program stack label (e.g. '4CP, ERS, LMP')
iso_id	uuid	FK	ISO administering this program (nullable if utility-run)
utility_id	uuid	FK	Utility administering this program (nullable if ISO-run)

Core Data Tables

SITE

One row per meter/account number. The central hub of the schema — every piece of equipment and enrollment links back to a site.

Column	Type	Key	Notes
site_id	uuid	PK	Unique identifier for each site/meter
client_id	uuid	FK	Owning client organization
region_id	uuid	FK	Geographic region of the site
utility_id	uuid	FK	Utility serving this site
site_name	varchar		Project or campus name (e.g. 'Main Campus', 'West Wing')
address	text		Full street address including city and state
account_number	varchar		Utility account number
meter_number	varchar		Meter identifier
energy_usage_kwh	float		Annual energy usage in kWh
avg_demand_kw	float		Average demand in kW
peak_demand_kw	float		Peak demand in kW
elec_cost_kwh	float		Electricity cost in $/kWh
gas_cost_mmbtu	float		Natural gas cost in $/MMBtu
bed_count	integer		Bed count for healthcare sites (nullable)
campus_sqft	integer		Conditioned campus size in sq ft (nullable)

SITE_ISO

Junction table linking sites to ISOs. Handles many-to-many relationships (a site may span multiple ISOs).

Column	Type	Key	Notes
site_id	uuid	FK	Reference to SITE — composite PK with iso_id
iso_id	uuid	FK	Reference to ISO — composite PK with site_id

EQUIPMENT

One row per asset enrolled or evaluated for a DR program. Replaces the wide RaaS/BESS/BMS/Solar column groups in the source spreadsheet.

Column	Type	Key	Notes
equipment_id	uuid	**PK**	Unique identifier for each piece of equipment
site_id	uuid	**FK**	Site where this equipment is located
equipment_type	varchar		Primary category: RaaS, BESS, BMS, Solar, Existing Backup Generation
equipment_subtype	varchar		Sub-type, e.g. HVAC, Steam Chillers, HPCH (for BMS)
nameplate_kw	float		Nameplate generation/capacity in kW
targeted_kw	float		Targeted dispatch capacity in kW
capacity_kwh	float		Energy storage capacity in kWh (BESS only, nullable)
included	boolean		Whether this equipment is included in the current proposal

ENROLLMENT

Links equipment to a DR program via a market participant. Holds all financial and project status data. One row per equipment-program pairing.

Column	Type	Key	Notes
enrollment_id	uuid	PK	Unique identifier for each enrollment
equipment_id	uuid	FK	Equipment being enrolled
program_id	uuid	FK	DR program the equipment is enrolled in
mp_id	uuid	FK	Market participant managing the enrollment
phase_internal	varchar		Internal project phase (e.g. 1, 2, Closed, Z)
phase_external	varchar		Client-facing phase label
mou_sent_date	date		Date MOU was sent to client
closed_date	date		Date project was closed/contracted
project_number	varchar		Internal project reference number
install_cost	numeric		Total installation cost in $
rebates	numeric		Applicable rebates in $ (BESS/Solar)
itc	numeric		Investment tax credit in $ (BESS/Solar)
annual_revenue	numeric		Gross annual revenue or savings in $/year
annual_mp_fee	numeric		Annual market participant fee in $/year
annual_fuel_cost	numeric		Annual fuel cost in $/year (generation assets)
annual_om	numeric		Annual operations and maintenance cost in $/year
net_revenue	numeric		Net annual revenue = revenue minus fees, fuel, and O&M;
payback_years	numeric		Simple payback period in years
npv	numeric		Net present value of the contract term in $
total_cash	numeric		Total considerations and cash over contract term in $
interval_data	boolean		Whether interval/AMI data has been obtained
notes	text		Free-text notes on the enrollment

SkySpark Integration Tables

These three tables connect the core schema to SkySpark without duplicating time-series data. SKYSPARK_REF is the bridge — it stores the point references the app uses to query SkySpark on demand. Time-series telemetry and environmental readings are fetched live from SkySpark and cached at the application layer; they are not stored in the database. DR_EVENT and EVENT_PERFORMANCE are the exception: event performance is derived/analyzed data produced by the app and written permanently to the database.

SKYSPARK_REF

Bridge table linking your schema to SkySpark. Stores the SkySpark point reference and site ID used by the app to query SkySpark's REST/Haystack API on demand. Can link to either SITE or EQUIPMENT (both FKs are nullable).

Column	Type	Key	Notes
ref_id	uuid	PK	Unique identifier for each SkySpark reference
site_id	uuid	FK	Linked site — nullable if reference is equipment-level
equipment_id	uuid	FK	Linked equipment — nullable if reference is site-level
skyspark_site_id	varchar		SkySpark internal site identifier
skyspark_point_ref	varchar		SkySpark point ref used to construct API queries
haystack_tags	text		Raw Haystack tag dict stored as JSON string
display_name	varchar		Human-readable label for this point
unit	varchar		Engineering unit (e.g. kW, degF, %)
active	boolean		Whether this reference is currently in use

DR_EVENT

One row per demand response dispatch. Links to ENROLLMENT so the program, equipment, and market participant are always traceable.

Column	Type	Key	Notes
event_id	uuid	PK	Unique identifier for each DR event
enrollment_id	uuid	FK	Enrollment that triggered this event
event_date	date		Calendar date of the event
start_ts	timestamp		Event start timestamp
end_ts	timestamp		Event end timestamp
event_type	varchar		e.g. Emergency, Economic, Test
status	varchar		e.g. Called, Completed, Excused, Missed
called_kw	float		Dispatch capacity requested in kW

EVENT_PERFORMANCE

One row per event per SkySpark point. Records baseline vs. actual performance for each asset during a DR event. baseline_method tracks the ISO-specific calculation used (e.g. 10-of-10, adjusted 10-of-10).

Column	Type	Key	Notes
perf_id	uuid	**PK**	Unique identifier for each performance record
event_id	uuid	**FK**	DR event this record belongs to
ref_id	uuid	**FK**	SkySpark point the data was sourced from
baseline_kw	float		Calculated baseline demand in kW
actual_kw	float		Actual demand during the event in kW
reduction_kw	float		Achieved reduction in kW (baseline minus actual)
reduction_pct	float		Reduction as a percentage of baseline
baseline_method	varchar		Baseline calculation method (e.g. 10-of-10, Adj 10-of-10)
passed	boolean		Whether the asset met its performance obligation
notes	text		Free-text notes on performance

- *Recommended database: PostgreSQL. Use UUID primary keys for all tables.*
- *Add indexes on commonly-filtered foreign key columns: client_id, iso_id, region_id, equipment_type, phase_internal.*
- *For time-series scale, consider TimescaleDB (a PostgreSQL extension) if event performance data grows large.*
- *Before migration, standardize lookup values — especially client_type and ISO strings — to populate reference tables with clean, canonical entries.*
- *SkySpark data (telemetry, environmental readings) is fetched live via the Haystack REST API using skyspark_point_ref; it is not stored in this database.*